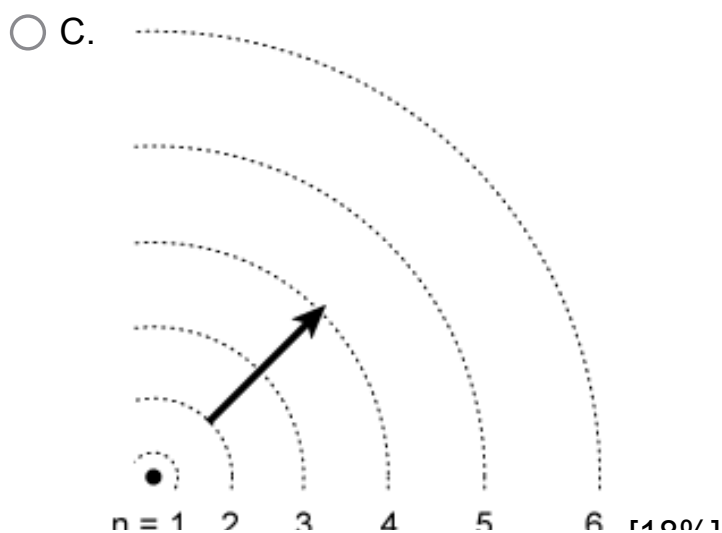
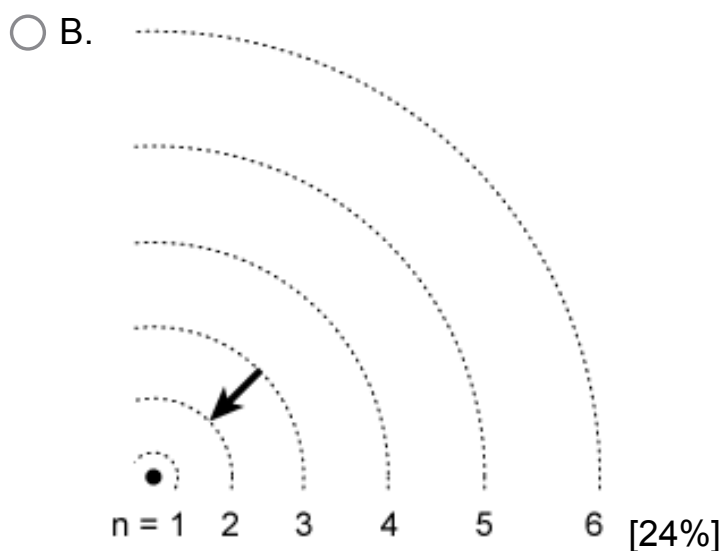
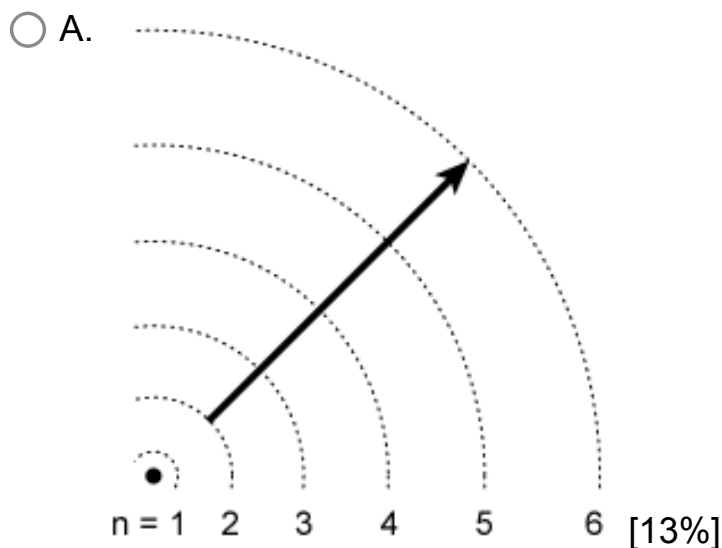


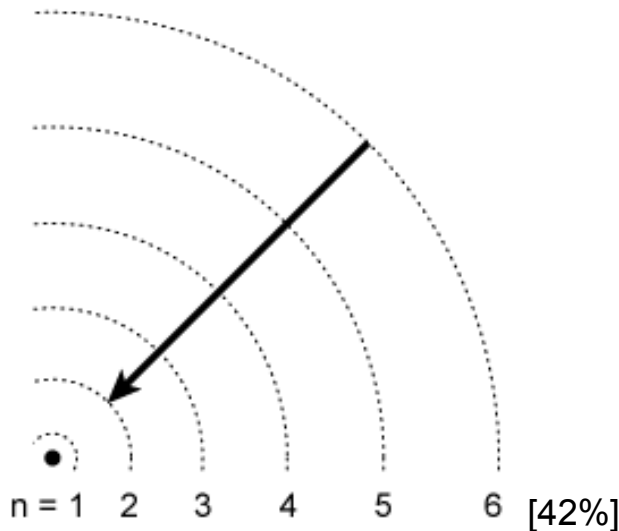
The emission line spectrum of hydrogen displays four spectral lines in the visible region of the spectrum.



According to the Bohr model of the atom, which of the diagrams below represents the spectral line at 410 nm in the emission line spectrum?



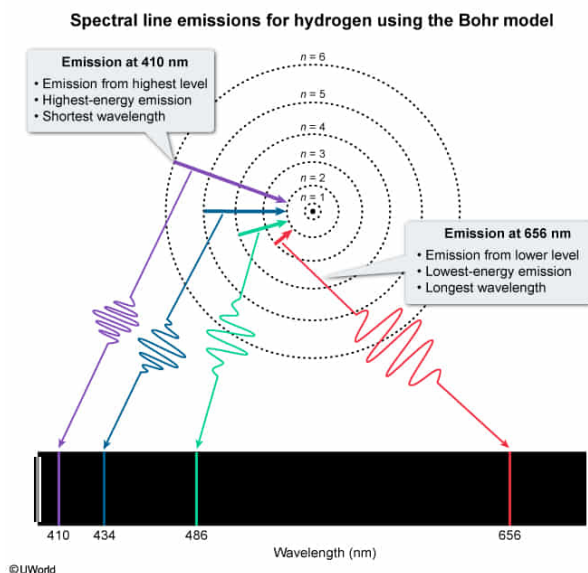
✓ ☒ D.



Correct

42% answered correctly

## Explanation:



In the **Bohr model** of the atom, several assumptions are made regarding electrons and their locations around the nucleus. The Bohr model asserts that:

- Electrons move around the nucleus in fixed **circular orbits**, which are only allowed at **particular intervals** from the nucleus.
- Electrons in orbits **farther from the nucleus** have **higher energy** than electrons in orbits closer to the nucleus.
- **Energy is absorbed** by an electron moving from a lower orbit to a higher orbit, but **energy is emitted** by an electron returning from a higher orbit back to a lower orbit.

- The energy that is absorbed or emitted by an electron equals the energy difference between two orbits.

According to the Bohr model, because electrons can exist only in certain orbits around the nucleus (not between these orbits), emissions from the hydrogen atom are seen only at particular wavelengths as a set of emission lines (a line spectrum) instead of as a continuous spectrum of all wavelengths. Likewise, in the [absorption spectrum](#), the same wavelengths are exclusively absorbed (seen as black lines).

Because [light energy](#) is inversely proportional to the wavelength, the line emission at 410 nm (the shortest wavelength in the line spectrum) must correspond to the electron originating in the highest-energy orbit.

Therefore, the energy diagram depicting the transition of an electron from  $n = 6 \rightarrow 2$  (the highest energy emission) is associated with the shortest spectral line wavelength (410 nm).

**(Choices A and C)** These diagrams show an *absorption* instead of an emission. Electrons that are excited from a lower to a higher energy level absorb, rather than emit, energy.

**(Choice B)** This diagram shows the emission associated with the spectral line at 656 nm. An electron moving from  $n = 3 \rightarrow 2$  will produce an emission with lower energy (longer wavelength) than an electron moving from  $n = 6 \rightarrow 2$ .

### Educational objective:

In the Bohr model of the atom, electrons move around the nucleus in fixed circular orbits at particular intervals. Electrons in orbits farther from the nucleus have higher energy. Energy equal to the difference between two orbits is absorbed by an electron moving to a higher orbit and is emitted by an electron moving to a lower orbit.

**Time spent:** 0 sec  
**Subject:**  
General Chemistry

**QID:** 402327  
**System:**  
Atomic Theory and Chemical Composition